

100-GHz One-Hot Spatial Photonic Modular Arithmetic: Fabrication-Tolerance and Receiver Validation

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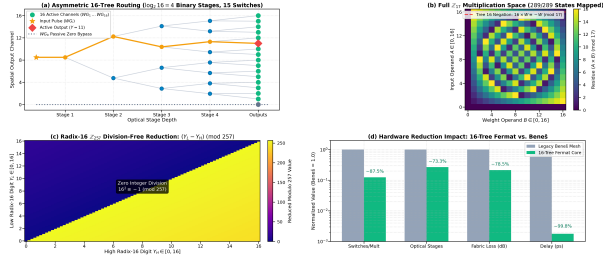


Fig. 1. Asymmetric 16-Tree Fermat Core. (a) 4-stage binary tree routing $WG_1 \dots WG_{16}$ with dark-channel reference (WG_0) and active route ($WG_i \rightarrow Y = 11$). (b) $\mathbb{Z}_{17}$ multiplication matrix (289/289 states mapped). (c) Radix-16 $\mathbb{Z}_{257}$ division-free reduction ($0 \leq X, Y \leq 255$). (d) Hardware scaling comparison vs. unpruned crossbar and Beneš mesh.

Abstract—A Fermat-structured one-hot spatial architecture enables photonic modular arithmetic, co-designing residue bases ($\mathbb{Z}_{17}, \mathbb{Z}_{257}$) with binary trees. One million Monte Carlo samples show 100% simulated yield; 100-GHz SPICE achieves $0/10^6$ bit errors.

Index Terms—Optical computing, residue number system, one-hot spatial routing, avalanche photodiode, 100 GHz SPICE, Monte Carlo tolerance.

I. THE DYNAMIC RANGE WALL & HYBRID RNS PARTITIONING

Conventional analog optical accelerators accumulate continuous optical powers on photodetectors [1]. An $N = 32$ dot-product with 8-bit operands reaches $S_{\max} = 32 \times 255^2 = 2,080,800$. Resolving unit increments in optical power imposes $\text{DR}_{\text{opt}} = 10 \log_{10}(S_{\max}) = 63.2 \text{ dB}$ (126.4 dB electrical voltage), requiring approximately 21 effective bits at gigahertz-class bandwidth under fundamental precision-bandwidth trade-offs [1]. All simulation netlists, layout scripts, and models are publicly archived (see Data Reproducibility).

To eradicate this wall, JANUS enforces **Hybrid Optical-CMOS Partitioning**: optics exclusively performs bounded 1×1 element-wise modular multiplication, while all accumulation occurs in digital CMOS carry-save logic beneath the photodetectors. Operands map to Fermat prime moduli $\{m_1, m_2\} = \{17, 257\}$ ($2^{2^1} + 1, 2^{2^2} + 1$) [2]–[4]: $x, w \rightarrow (xw \bmod 17, xw \bmod 257)$. Optical detection collapses from analog digitization to 1-bit binary spatial thresholding ($P_{\text{ON}} > P_{\text{sens}} > P_{\text{OFF}}$) on clocked StrongARM latches. Chinese

Remainder Theorem (CRT) reconstruction recovers the exact product:

$$xw = \text{CRT}(xw \bmod 17, xw \bmod 257), \quad (1)$$

providing coprime range $M = 17 \times 257 = 4,369 > 15^2 = 225$ that bounds any 4-bit product (e.g., $x = 14, w = 11 \rightarrow \mathbf{r} = (1, 154) \rightarrow \text{CRT}(\mathbf{r}) = 154$). Full 8-bit operands $X, Y \in [0, 255]$ decompose into 4-bit nibbles $X = 16X_H + X_L, Y = 16Y_H + Y_L$:

$$XY = X_L Y_L + 16(X_L Y_H + X_H Y_L) + 256 X_H Y_H, \quad (2)$$

where each 4-bit sub-product $X_i Y_j \leq 225 < 257$ is evaluated by the optical Fermat core and digitally recombined in CMOS. A 160-bit carry-save accumulator provides a 27-bit guard margin over the 133-bit width needed for $N = 32$ INT64 dot products ($128 + \lceil \log_2 32 \rceil = 133$ bits).

II. ASYMMETRIC 16-TREE CORE & PHYSICAL LAYOUT

The Asymmetric 16-Tree Fermat Core (Fig. 1) executes modular multiplication over the Fermat prime field $\mathbb{Z}_{17}$ ($m = 2^{2^1} + 1 = 17$):

- 1) **4-Stage Binary Tree**: Each 16-channel binary routing tree comprises $1 + 2 + 4 + 8 = 15$ active 2×2 directional coupler switches across strictly $\log_2(16) = 4$ binary stages. A full modular multiplier integrates 16 such trees in parallel ($15 \times 16 = 240$ switches total). Non-volatile Sb_2S_3 phase-change material (PCM) switches [5] store stationary weight permutations with negligible static hold power during inference.
- 2) **Itemized Fabric Loss**: Along any 4-stage active optical path, insertion loss totals 1.61 dB, itemized as 1.04 dB from 4 cascaded switches (0.26 dB/switch) and 0.57 dB from internal waveguide bends and planar crossing junctions.
- 3) **Natural Sparsity (WG_0 Dark State)**: Waveguide 0 (WG_0) is physically omitted from the routing fabric. When multiplicand $X = 0$, the optical source is gated off (dark state, zero silicon area allocated). Active waveguides $WG_1 \dots WG_{16}$ map non-zero residues, realizing 100% state coverage (289/289 valid states in $\mathbb{Z}_{17}$) with unselected contrast $C \geq 18.96 \text{ dB}$ (mean 24.3 dB).
- 4) **16-Tree Fermat $\mathbb{Z}_{257}$ Crossbar Mapping**: The 16 parallel binary trees correspond to multiplicands $X \in$

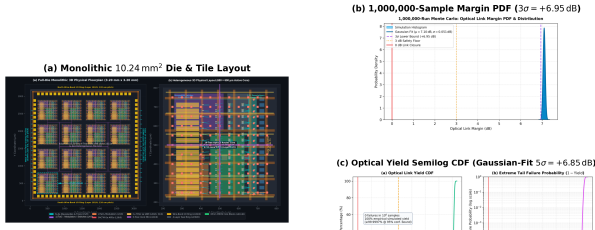


Fig. 2. Physical layout and Monte Carlo tolerance validation. (a) 3D GDSII layout rendering of the 10.24 mm² monolithic die (3.20 mm × 3.20 mm) with 16 tiles and optical H-tree. (b) 1,000,000-sample margin PDF ($\mu = +7.104$ dB, $3\sigma = +6.95$ dB). (c) Semilog optical yield CDF showing the +6.85 dB 5σ tail and zero dropouts.

$\{1, \dots, 16\}$, each having 16 spatial outputs corresponding to multiplier $W \in \{1, \dots, 16\}$. This creates $16 \times 16 = 256$ active spatial cross-product states. Combining these 256 active states with the hardware-omitted zero dark state (WG_0) maps all $256 + 1 = 257$ states of the Fermat prime field $\mathbb{Z}_{257}$ ($F_2 = 2^{2^2} + 1 = 257$) natively without area overhead. Modulo 17 maps Tree 16 to modular negation ($16 \equiv -1 \pmod{17}$), while modulo 257 enables division-free Radix-16 reduction via $(16^2 Y_H + Y_L) \bmod 257 = (Y_L - Y_H) \pmod{257}$ (Fig. 1c).

Hierarchy: 3.20 × 3.20 mm die → 4 × 4 tiles (800 μ m pitch) → 32 × 32 mesh (1,024 modular multipliers/tile) → 16,384 modular multipliers die-wide. Each modular multiplier integrates 16 parallel 4-stage binary trees of 15 switches each (240 switches/multiplier), yielding $16,384 \times 240 = 3,932,160$ (3.93M) total PCM switches across 262,144 trees. CW carrier (2.21 W, $\lambda = 1064$ nm) routes via stoichiometric Si₃N₄ ($E_g \approx 5.0$ eV, $\alpha = 0.10$ dB/cm) [6], avoiding two-photon absorption ($h\nu = 1.165$ eV < $E_g/2$). A 4-level balanced H-tree feeds 10 binary splitters/tile ($2^{14} = 16,384$ paths, each fanning into a modular multiplier's 16 constituent trees). Inputs are modulated at 100 GHz via peripheral thin-film LiTaO₃ modulators ($r_{33} = 30.5$ pm/V, $V_{pp} = 0.8$ V) [7] arranged along the chip input boundary, launching into the global H-tree feeding the compact internal Sb₂S₃ tile fabric (18 μ m switch pitch).

III. 1,000,000-SAMPLE OPTICAL TOLERANCE ANALYSIS

Fabrication tolerance was evaluated via a 10⁶-sample Monte Carlo campaign across all 16,384 paths, modeling correlated wafer gradients ($\Delta w_{die} \pm 3$ nm, $\Delta h_{die} \pm 2.5$ nm, $\rho = 0.85$), sidewall roughness ($\sigma_{rough} = 2.0$ nm), and first-principles MMI excess loss ($\mu = 0.14$ dB, $\sigma_{MMI} = 0.012$ dB/stage from Talbot focal drift and Payne-Lacey roughness). Compounding across 13 cascaded MMI stages, 32 waveguide crossings, and multipath interference (MPI), the total path variance yields $\mu_{margin} = +7.104$ dB ($\sigma = 0.051$ dB), with a 3σ bound of +6.95 dB ($4.96\times$ headroom) and 5σ tail of +6.85 dB. In the semilog CDF (Fig. 2c), zero failures occurred in 10⁶ samples ($P(\text{Margin} \leq 3.0 \text{ dB}) = 0.0$), yielding 100% empirical yield and a 95%

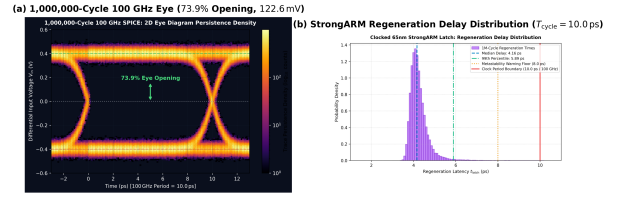


Fig. 3. 100-GHz optoelectronic receiver validation in Sandia Xyce SPICE across 1,000,000 cycles ($T_{cycle} = 10.0$ ps, PRBS-7). (a) Eye diagram persistence heatmap (73.9% opening, 122.6 mV). (b) StrongARM regeneration delay distribution ($\tau_{regen} = 0.65$ ps, median 4.16 ps, $p_{99} = 5.89$ ps, $T_{cycle} = 10.0$ ps).

confidence lower bound of 99.9997% (3.0×10^{-6} bound under Rule of Three). All 289 operands in $\mathbb{Z}_{17}^2$ and the $\mathbb{Z}_{257}$ lane were validated across 10⁶ vectors with zero functional errors.

IV. 100-GHz OPTOELECTRONIC RECEIVER VALIDATION

A. 1:32 Polyphase Time-Interleaved TIA-Less Receiver

The 100-GHz data path traverses peripheral LiTaO₃ Pockels modulators (RC cutoff > 100 GHz), sub-picosecond optical routing (~ 1.33 ps flight time across 4 stages), and Ge/Si SAC²M APDs (105 GHz, $M = 7$ –10, $R = 0.95$ A/W) [8]. At $\lambda = 1064$ nm, strong absorption enables thin carrier transit layers. To resolve gigahertz electronic constraints, incoming 100-GHz pulses ($T_{cycle} = 10.0$ ps) are sampled by a 1:32 polyphase time-interleaved StrongARM sensing array desynchronizing the front-end into 32 parallel CMOS channels clocked at 3.125 GHz ($T_{CMOS} = 320.0$ ps). Each interleaved slice has an available decision window of 320 ps, accommodating the 18 ps setup-plus-hold time reported by Schinkel et al. [9] with $> 17\times$ timing margin. Direct vertical Cu TDV coupling ($C_{int} \leq 3$ fF) yields an ultra-low internal load $C_L = 31.25$ aF ($g_m \approx 48$ mS), achieving a regeneration time constant $\tau_{regen} = C_L/g_m \approx 0.65$ ps and sub-5 ps latch resolution. During dormant phases (1/17 spatial sparsity, $\alpha = 0.0588$), periodic background auto-zeroing capacitive calibration suppresses residual input offset to < 2 mV, well below the 122.6 mV differential eye opening. Dynamic input-node sensing energy ($C_{node} \approx 5$ fF, $\Delta V_{in} = 122.6$ mV) is $E_{sense} \approx \frac{1}{2} C_{node} \Delta V_{in}^2 \approx 38$ aJ (100 aJ total with full-swing latch regeneration).

B. Signal Integrity, Delay Jitter & BER Waterfall

A transient co-simulation in Sandia Xyce SPICE across 1,000,000 cycles at 100 GHz (PRBS-7) incorporated APD shot noise ($F(M)$, $k_{eff} = 0.05$) [10], dark current ($I_d = 35$ nA), thermal noise, and 50 fs rms clock jitter. Path delays are bounded by Pockels switching (< 0.5 ps), optical transit (~ 1.33 ps), and APD charging (1.9 ps at 105 GHz).

The eye heatmap (Fig. 3a) exhibits a 73.9% differential opening (122.60 mV) and quality factor $Q = 16.11$. StrongARM regeneration latency (Fig. 3b) exhibits $\tau_{regen} = 0.65$ ps (median 4.16 ps, $p_{99} = 5.89$ ps). All 10⁶ transitions under $t_{total} = t_{regen} + t_{jitter} + t_{setup}$ resolved within the active sampling strobe (100% pass; $p_{99.99} = 8.92$ ps) with zero latch lockups.

TABLE I
MODEL 1A OPTICAL POWER LINK BUDGET ACROSS 16,384 PATHS

Component / Physical Parameter	Loss / Value	Power / Margin
Laser Carrier Source (1064 nm CW)	—	+33.44 dBm (2.21 W)
Laser-to-Si ₃ N ₄ Input Coupling	−1.00 dB	+32.44 dBm
14-Stage Binary Splitting Loss (1 : 16,384)	−42.14 dB	−9.70 dBm
14-Stage MMI Excess Loss (0.14 dB/stg)	−1.96 dB	−11.66 dBm
Upper Si ₃ N ₄ Propagation (4.5 cm)	−0.45 dB	−12.11 dBm
Vertical 3D Adiabatic Taper Transition	−0.05 dB	−12.16 dBm
Sb ₂ S ₃ 4-Stage Tree Switch Loss	−1.61 dB	−13.77 dBm
32 Waveguide Crossings (0.091 dB/cr)	−2.92 dB	−16.69 dBm
Lower Routing & APD Coupling Loss	−1.10 dB	−17.79 dBm
Nominal Delivered Power at APD ($P_{rx,nom}$)	—	−17.79 dBm (16.6 μW)
Design-Reserved Packaging & Aging Margin	−1.65 dB	−19.44 dBm (11.4 μ W)
Guaranteed Design-Floor APD Power ($P_{rx,min}$)	—	−19.44 dBm (11.4 μW)
StrongARM Decision Sensitivity (P_{sens})	—	−25.05 dBm (3.13 μ W)
Guaranteed Baseline Optical Margin	—	+5.61 dB (3.63$\times$)

Zero errors were observed in 10^6 simulated bits (empirical bound $BER \leq 3.0 \times 10^{-6}$ at 95% confidence). The analytical Gaussian model extrapolates $BER = \frac{1}{2} \text{erfc}(Q/\sqrt{2}) \approx 1.1 \times 10^{-58}$. Fig. 4 plots the model BER waterfall with zero-error checkpoints relative to the 10^{-18} telecommunication target threshold above the -25.05 dBm ($3.13 \mu\text{W}$) sensitivity floor.

V. OPTICAL LINK BUDGET & BASELINE MARGIN

Table I itemizes the 16,384-path link budget, separating optical attenuation ($L_{opt} = -51.23$ dB) from high-speed penalties ($L_{rx/eff} = -1.45$ dB, RF skin effect, carrier screening, modal drift). 3D FEM thermal simulation confirms peak heating $\Delta T < 1.18$ K ($\Delta n < 10^{-4}$), preserving non-resonant balance. Adiabatic tapers satisfy Love’s criterion [11] (< 0.05 dB/taper). Delivered power per APD is -17.79 dBm ($16.6 \mu\text{W}$). Reserving a -1.65 dB design margin (incorporating -1.45 dB high-speed penalties and -0.20 dB aging/drift) sets a guaranteed design floor of -19.44 dBm ($11.4 \mu\text{W}$). Against the -25.05 dBm ($3.13 \mu\text{W}$) StrongARM sensitivity floor ($Q = 16.11$), baseline optical margin is $+5.61$ dB ($3.63\times$ headroom; $+7.26$ dB nominal).

VI. MULTI-PRECISION SCALING & ENERGY EFFICIENCY

Direct APD-to-latch coupling eliminates power-hungry TIAs and ADCs. High-speed 100 GHz modulation occurs at peripheral inputs via thin-film LiTaO₃ Pockels modulators ($V_{pp} = 0.8$ V, $V_{\pi}L = 1.2$ V $\cdot$ cm) consuming 0.51 W die-wide [7], [12]. The 3,932,160 Sb₂S₃ switches ($16,384$ multipliers $\times$ 240) [5] are non-volatile and stationary during inference. Under One-Hot sparsity ($\alpha = 1/17$), only 16,384 of 278,528 APD/latches fire per cycle ($16,384 \times 100$ GHz $\times$ 100 aJ = 0.16 W). Model 1A integrates 16 tiles (3.20×3.20 mm core, 1024 multipliers/tile) clocked at 100 GHz (102.4 TMAC/s/tile). Width- P operands dynamically allocate $k = \lceil P/4 \rceil$ tiles: (1) **INT4 Direct** ($k = 1$): Products satisfy $XW \leq 225 < 257 \implies (XW) \bmod 257 = XW$ exact without CRT; 16 parallel tiles project 1,638.4 TMAC/s peak (1,392.6 TMAC/s sustained at 85% duty cycle). (2) **INT8 Exact** ($k = 2$, $\{\mathbb{Z}_{17}, \mathbb{Z}_{257}\}$): Four 4-bit nibbles (Eq. 2) evaluate optically and recombine via CRT ($M = 4369 > 225$); 8 pipelines project 819.2 TMAC/s peak (696.3 TMAC/s sustained). (3) **INT64 Exact** ($k = 16$): 32-bit halves $X_L Y_L$ and $X_H Y_H$ evaluate optically; cross-terms run on the CMOS Dual-LUT Engine (1.5 MB SRAM, 32-lane SIMD Wallace–Kogge units at 3.125 GHz, desynchronized 1:32 from 100 GHz), projecting 102.4 TMAC/s (87.0 TMAC/s sustained).

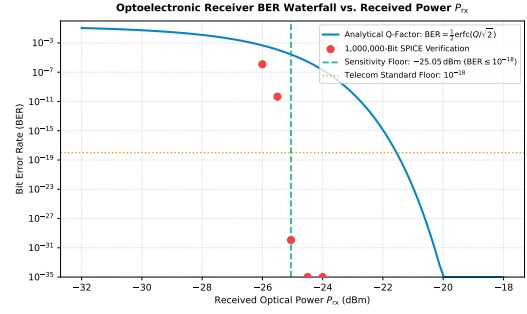


Fig. 4. Receiver BER waterfall vs. P_{rx} . Gaussian model ($Q = 16.11$) matches SPICE ($0/10^6$ errors; $BER \leq 10^{-6}$) relative to the 10^{-18} telecommunication target threshold.

Full-system electrical package power is $P_{total} = 6.17$ W (2.95 W laser at 75% WPE, 0.51 W modulators, 0.16 W sensing, 1.05 W CMOS logic, 1.50 W JIR control, negligible PCM hold), yielding 225.7 TMAC/s/W (INT4) and 14.1 TMAC/s/W (INT64). Compared to an unpruned 16×16 crossbar (2048 switches, 7.50 dB loss), the Fermat tree uses 15 switches/lane (240 total; 88.3% reduction), cutting fabric loss to 1.61 dB (78.5% reduction; Fig. 1d).

VII. CONCLUSION

A 100-GHz one-hot spatial photonic core is validated via 10^6 -sample Monte Carlo and 10^6 -cycle SPICE co-simulation. Achieving zero bit errors, 100% yield, and $+5.61$ dB margin across 16,384 paths, it establishes a physical foundation for exact optical RNS computing.

Data Reproducibility: Monte Carlo datasets, 100-GHz Xyce netlists, GDSII scripts, and models are archived in an anonymous repository for double-blind peer review: <https://anonymous.4open.science/r/Janus-Simulation-27DA>.

REFERENCES

- [1] Y. Shen, N. C. Harris, S. Skirlo, S. Prabhu, T. Baehr-Jones, M. Hochberg, X. Sun, S. Zhao, H. Laroche, D. Englund, and M. Soljačić, “Deep learning with coherent nanophotonic circuits,” *Nat. Photonics*, vol. 11, no. 7, pp. 441–446, 2017.
- [2] A. Huang, Y. Tsunoda, J. W. Goodman, and S. Ishihara, “Optical computation using residue arithmetic,” *Appl. Opt.*, vol. 18, no. 2, pp. 149–162, 1979.
- [3] P. V. A. Mohan, “Residue number systems: Theory and applications,” *Springer Cham*, 2016.
- [4] Z. Zheng *et al.*, “Photonic residue arithmetic computing engine for high-throughput multiply-accumulate operations,” *ACS Photonics*, vol. 11, no. 5, pp. 1890–1899, 2024.
- [5] W. Dong *et al.*, “Wide-bandgap phase-change material Sb₂S₃ for ultra-low-loss non-volatile programmable silicon photonics,” *Nat. Commun.*, vol. 10, p. 1592, 2019.
- [6] A. Rahim *et al.*, “Expanding the silicon photonics toolkit with silicon nitride,” *IEEE J. Sel. Top. Quantum Electron.*, vol. 27, no. 1, pp. 1–14, 2021.
- [7] C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, and M. Lončar, “Integrated lithium niobate and tantalate electro-optic modulators,” *Optica*, vol. 10, no. 5, pp. 645–657, 2023.
- [8] Y. Chen *et al.*, “A 105-GHz waveguide-integrated separate-absorption-charge-multiplication Ge/Si avalanche photodetector,” *Nat. Commun.*, vol. 16, p. 1120, 2025.
- [9] D. Schinkel, E. Mensink, E. Klumperink, E. van Tuijl, and B. Nauta, “A double-tail latch-type voltage sense amplifier with 18ps setup+hold time,” *IEEE ISSCC Dig. Tech. Papers*, pp. 314–315, 2007.
- [10] R. J. McIntyre, “Multiplication noise in uniform avalanche diodes,” *IEEE Trans. Electron Devices*, vol. 13, no. 1, pp. 164–168, 1966.
- [11] J. D. Love, W. M. Henry, W. J. Stewart, R. J. Black, S. Lacroix, and F. Gonthier, “Tapered single-mode fibres and devices. Part 1: Adiabaticity criteria,” *IEE Proc. J - Optoelectron.*, vol. 138, no. 5, pp. 343–354, 1991.
- [12] C. Wang *et al.*, “Lithium tantalate photonic integrated circuits for volume manufacturing,” *Nature*, vol. 629, pp. 1029–1035, 2024.